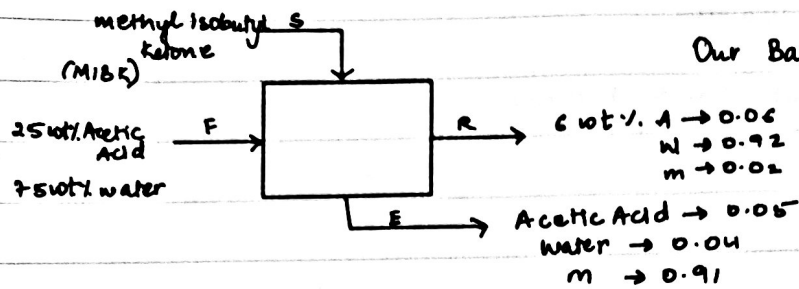


Homework 9 210

RAINA SWAMY

1.



Our Basis is 100 kg/h feed.

$$R_W = \frac{6\% - 2.85\%}{11.7\% - 2.85\%} (85.8\% - 95.46\%) + 95.46\%$$

$$= 92.02\%$$

$$R_m = \frac{6\% - 2.85\%}{11.7\% - 2.85\%} (2.5\text{wt}\% - 1.7\text{wt}\%) + 1.7\%$$

$$R_A = 6\text{wt}\% \quad E_A = 5\text{wt}\%$$

$$E_W = \frac{5\% - 1.87\%}{8.9\% - 1.87\%} (5.4\% - 2.80\%) + 2.80\%$$

$$= 3.96\% \approx 0.04$$

$$E_m = \frac{5\% - 1.87\%}{8.9\% - 1.87\%} (85.7\% - 95.7\%) + 95.7\%$$

$$= 91.25\% \approx 0.91$$

1) 25

2) 25

3) 25

4) 25

$$\frac{100}{100}$$

$$A: 0.06 \dot{m}_R + 0.05 \dot{m}_E = 25 \text{ kg/h}$$

$$W: 0.92 \dot{m}_R + 0.04 \dot{m}_E = 75 \text{ kg/h}$$

$$m: 0.02 \dot{m}_R + 0.91 \dot{m}_E = \dot{m}_S$$

$$\dot{m}_E = 428.2 \text{ kg/h}$$

$$\dot{m}_R = 62.9 \text{ kg/h}$$

$$\dot{m}_S = 390.9 \text{ kg/h}$$

$$\dot{m}_F + \dot{m}_S = \dot{m}_R + \dot{m}_E$$

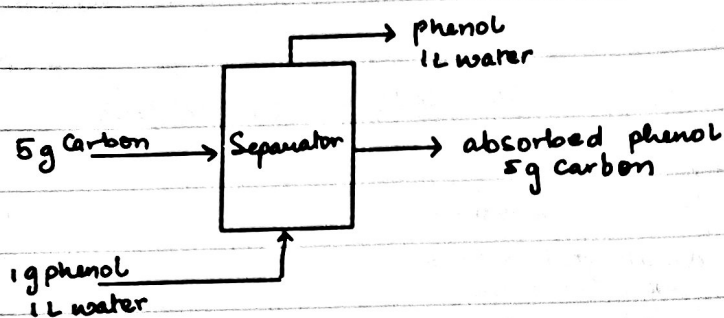
$$100 \text{ kg/h} + 390.9 \text{ kg/h} = 428.2 \text{ kg/h} + 62.9 \text{ kg/h}$$

$$490.9 \text{ kg/h} \approx 491.1 \text{ kg/h}$$

$$\frac{25}{25}$$

∴ 390.9 kg solvent for 100 kg feed

2.



$$1\text{g phenol } [\text{C}_6\text{H}_5\text{OH}] = 10.638 \text{ mmol}$$

$$C_p = \text{phenol conc. in liquid} \quad (\text{mmol/L})$$

$$q_p = \text{phenol absorbed on carbon} \quad (\text{mmol/g})$$

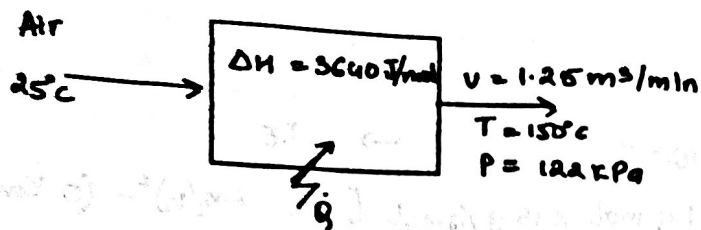
$$\text{Material balance equation:} \quad 10.638 = C_p(1\text{L}) + q_p(5\text{g})$$

Trial and error to see the point that satisfies the equation

$$C_p \sim 0.6 \text{ mmol/L} \quad \& \quad q_p = 2 \text{ mmol/g}$$

$$\% \text{ removal} = \frac{2 \times 5}{10.638} \times 100\% = \boxed{95\%}$$

$$\frac{25}{25}$$



Energy Balance:

$$\Delta H + \cancel{\Delta PE} + \cancel{\Delta KE} = Q - \cancel{W_s}$$

$$\therefore \Delta H = Q$$

$$\therefore Q = 3640 \text{ J/mol}$$

$$3640 \frac{\text{J}}{\text{mol}} \cdot n_{\text{air}} \frac{\text{mol}}{\text{s}} \cdot \frac{\text{kJ}}{1000 \text{ J}}$$

$$P = 122 \text{ kPa} = 122000 \text{ Pa}$$

$$V = 1.25 \frac{\text{m}^3}{\text{min}} \cdot \frac{1 \text{ min}}{60 \text{ s}} = 0.0208 \frac{\text{m}^3}{\text{sec}}$$

$$R = 8.314 \text{ m}^3 \cdot \text{Pa} / \text{mol} \cdot \text{K}$$

$$T = 423 \text{ K}$$

$$PV = nRT$$

$$n = \frac{PV}{RT} = \frac{(122000 \text{ Pa})(0.0208 \text{ m}^3/\text{sec})}{(8.314 \text{ m}^3 \cdot \text{Pa} / \text{mol} \cdot \text{K})(423 \text{ K})} = \frac{2537.6}{3516.82} = 0.721 \frac{\text{mol}}{\text{s}}$$

$$\therefore Q = 3640 \frac{\text{J}}{\text{mol}} \cdot 0.721 \frac{\text{mol}}{\text{s}} \cdot \frac{\text{kJ}}{1000 \text{ J}}$$

$$\boxed{Q = 2.624 \text{ kW}}$$

b. more information is required, we need to know the inlet velocity nor the diameters of the inlet and outlet

$$\text{This is based on } \Delta E_k = \frac{1}{2} u_1^2 - u_2^2$$

25/25

4. 1 g mol H_2O

a) decelerating from 0 to 100 km/h $\rightarrow KE$

$$\Delta E_K = \frac{m}{2} (v_2^2 - v_1^2) = \frac{1 \text{ g mol} \times 18 \text{ g/gmol}}{2} [(100 \text{ km/h})^2 - (0 \text{ km/h})^2]$$

$$= 9000 \frac{\text{g km}^2}{\text{hr}^2} \cdot \frac{1 \text{ kg}}{1000 \text{ g}} \cdot \frac{10^3 \text{ m}^2}{\text{km}^2} \cdot \frac{\text{h}^2}{3600^2 \text{ s}^2} \cdot \frac{\text{J}}{\text{kg m}^2/\text{s}^2}$$

$$= 6.9 \text{ J} \quad \checkmark$$

b. Rising 100 m $\rightarrow PE$

$$\Delta E_p = mg(h_2 - h_1) = (18 \text{ g})(9.8 \text{ m/s}^2)(100 \text{ m} - 0 \text{ m})$$

$$= 17651 \frac{\text{g m}^2}{\text{s}^2} \cdot \frac{1 \text{ kg}}{1000 \text{ g}} \cdot \frac{\text{J}}{\text{kg m}^2/\text{s}^2}$$

$$= 17.6 \text{ J} \quad \checkmark$$

c. Pressure $\uparrow$ at 100°C from 0.1 bar to 1 bar $\rightarrow U$

$$\Delta U = m \Delta \hat{U}$$

$$\Delta \hat{U} = \Delta H$$

$$U = 18 \text{ g} \cdot (2506.2 - 2515.5) \text{ J/g}$$

$$= -167 \text{ J} \quad \checkmark$$

d. $\uparrow$ Temp from 0 to 100°C at 5 bar

$$\Delta U = n C_v \Delta T = 1 \text{ g mol} \cdot 75.4 \frac{\text{J}}{\text{g mol} \cdot ^\circ\text{C}} \cdot (100 - 0^\circ\text{C})$$

$$\Delta U = 7540 \text{ J} \quad \checkmark$$

e. $\Delta \hat{U}_r = \Delta \hat{H}_r - RT \Delta n_r = \left(285.84 \frac{\text{kJ}}{\text{gmol}} \right) \left(\frac{1000 \text{ J}}{\text{kJ}} \right) - \left(8.3144 \frac{\text{J}}{\text{gmol} \cdot \text{K}} \right) (299 \text{ K})(0.5)$

$$= 284,600 \text{ J/gmol}$$

$$\Delta U_r = n \Delta \hat{U}_r = 284,600 \text{ J} \quad \checkmark$$

f. $\Delta E = (\Delta m) c^2 = 18 \text{ g} (2.99 \times 10^8 \text{ m/s})^2 \cdot \frac{1 \text{ kg}}{1000 \text{ g}} \cdot \frac{1 \text{ J}}{1 \text{ kg m}^2/\text{s}^2} = 1.6 \times 10^6 \text{ J} \quad \checkmark$